

Homework (Carolina Reaper)

- Q1.** A music producer is working with a quadratic equation $x^2 + 6x + 8$ to generate a rhythmic pattern. What is the vertex form of this equation?
- (A) $(x + 3)^2 - 1$
 (B) $(x + 3)^2 + 1$
 (C) $(x + 4)^2 - 8$
 (D) $(x + 2)^2 + 4$
 (E) $(x - 3)^2 - 1$
- Q2.** To create a canvas with a quadratic shape, an artist uses the equation $x^2 - 4x - 3$. What is the vertex of this parabola?
- (A) $(2, -7)$
 (B) $(-2, 7)$
 (C) $(2, -1)$
 (D) $(-1, 2)$
 (E) $(0, 0)$
- Q3.** A sound wave follows the quadratic equation $x^2 + 2x + 1 = 0$. Which of the following is the factored form of this equation?
- (A) $(x + 1)(x - 1)$
 (B) $(x + 1)^2$
 (C) $(x - 1)^2$
 (D) $(x + 2)(x + 1)$
 (E) $(x - 2)(x + 1)$
- Q4.** A painter is using a quadratic equation $x^2 - 2x - 6$ to create a unique pattern. What is the value of x that minimizes this function?
- (A) -1
 (B) 1
 (C) 2
 (D) -3
 (E) 3
- Q5.** To generate a specific audio frequency, a quadratic equation $x^2 + 8x + 15$ is used. What is the vertex form of this equation?
- (A) $(x + 4)^2 - 1$
 (B) $(x + 4)^2 + 1$
 (C) $(x + 3)^2 - 6$
 (D) $(x + 2)^2 + 7$
 (E) $(x + 4)^2 - 1$
- Q6.** A musician is working with the quadratic equation $x^2 - 10x + 24$. What is the factored form of this equation?
- (A) $(x - 4)(x - 6)$
 (B) $(x + 4)(x - 6)$
 (C) $(x - 2)(x - 12)$
 (D) $(x + 2)(x + 12)$
 (E) $(x - 8)(x - 3)$
- Q7.** An artist is using the quadratic equation $x^2 + 12x + 36$ to create a geometric pattern. What is the vertex of this parabola?
- (A) $(-6, 0)$
 (B) $(-6, -36)$
 (C) $(-6, 36)$
 (D) $(-12, 0)$
 (E) $(-12, -36)$
- Q8.** A music composer uses the quadratic equation $x^2 - 8x + 12$. What is the value of x that maximizes this function?
- (A) -2
 (B) 2
 (C) 4
 (D) 6
 (E) 8

Open Ended Questions (FRQ)

- Q9.** A painter is using a quadratic equation $x^2 - 6x + 8$ to create a mural. Convert this equation into vertex form and find the vertex of the parabola. Show all work and explain your reasoning.
- Q10.** A sound engineer is working with a quadratic equation $x^2 + 4x + 4$. Complete the square to rewrite this equation in vertex form. Then, find the x -intercepts and the y -intercept of the parabola. Provide a clear explanation of each step.

Chef's Tips

- Ensure students show all work and explain their reasoning for FRQs.
- Allow students to use calculators for checking answers only.
- Encourage students to use graphing tools to visualize the parabolas.